TELECOM CUSTOMER CHURN ANALYSIS

Predictive Modelling · Customer Segmentation · Retention Strategy

ELEVATE LABS INTERNSHIP PROJECT | DATA ANALYTICS DOMAIN | 2026

7,043	26.5%	0.848	976
Total Customers	Churn Rate	Best ROC-AUC	At-Risk Customers

Dataset: **IBM Telco Customer Churn** Rows: 7,043 Features: 21 (+7 engineered) Target: Churn (Yes/No) Source: Kaggle / IBM

OVERVIEW — Abstract, Introduction & Business Question

This project analyses the IBM Telco Customer Churn dataset (7,043 customers, 21 features) to uncover the root causes of customer attrition and build a predictive model that flags at-risk customers before they leave. Churn costs roughly 5× more than retention, and this dataset's 26.5% churn rate means more than one in four customers leaves every billing cycle. Four classification models were benchmarked — Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting — with Logistic Regression achieving the best ROC-AUC of 0.848 (CV-AUC 0.842). Every customer was scored with `predict_proba()` and grouped into three actionable segments — At Risk, Dormant, and Loyal — each paired with a targeted retention strategy, turning raw billing and usage data into a retention playbook a real team can act on immediately. **Business question:** which customers are most likely to churn, why are they leaving, and what specific interventions will retain them at the lowest possible cost?

KEY EDA INSIGHTS — 9 Charts Produced, Top Drivers Below

- **Contract type is the #1 churn driver —** month-to-month customers churn at **42.7%** vs just **2.8%** for two-year contracts, a 15× gap.
- **Early tenure is highest risk —** customers in their first 12 months churn at **47.7%**; churned customers average 18 months tenure vs 38 for retained.
- **Price sensitivity is real —** churned customers pay **$74.44/month** on average vs $61.27 for retained customers.
- **Service type matters —** fiber optic users churn at **41.9%** vs only 7.4% for customers with no internet service.
- **Payment method is a signal —** electronic check payers churn at **45.3%**, the highest of any payment method.
- **Correlation confirms it —** tenure has the strongest numeric correlation with churn (–0.352); senior citizens churn at 41.7%.

Chart 2 — Contract Type: The #1 Churn Driver

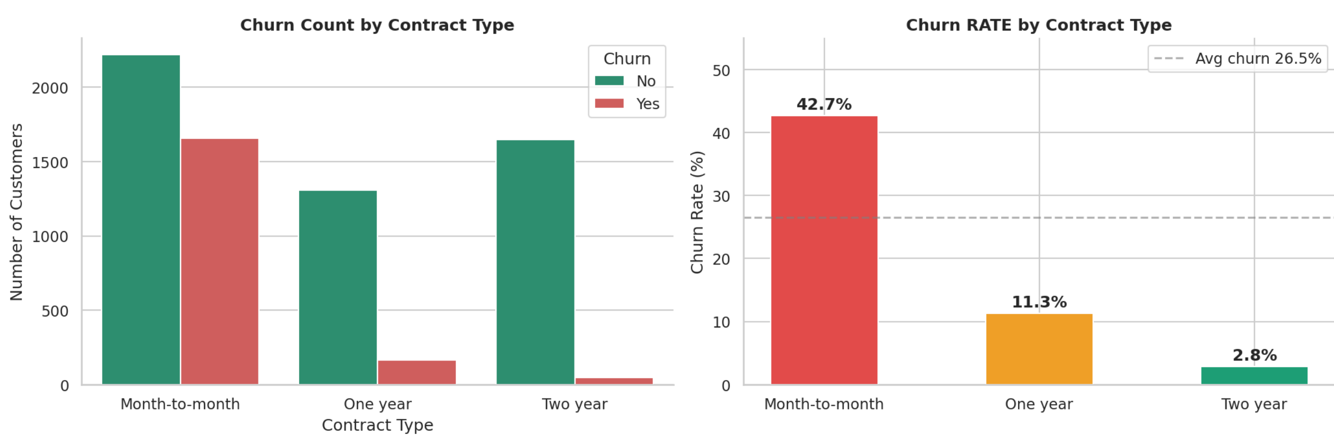

Chart — Contract Type: The #1 Churn Driver (month-to-month vs one-year vs two-year)

TOOLS & METHODS

Python 3.10 · Pandas · NumPy · Matplotlib / Seaborn · Scikit-learn (models, scaling, metrics) · LabelEncoder & One-Hot Encoding · SMOTE (class balancing) · Permutation Importance (model-agnostic explainability) · Joblib (model persistence) · ReportLab (this report)

MODEL BUILDING & EVALUATION — Preprocessing, Feature Engineering & 4 Models Benchmarked

Preprocessing pipeline: fixed TotalCharges type and 11 missing values, dropped customerID, label-encoded 6 binary columns, one-hot encoded Contract / InternetService / PaymentMethod, engineered 5 new features (AvgMonthlySpend, tenure_group, TotalServices, HasStreaming, HasProtection), scaled numeric features, and used an 80/20 stratified split (5,634 / 1,409) with **SMOTE** to balance the 73:27 class split to 50:50 for training — 28 features total. Models were evaluated on the original, unbalanced test set; ROC-AUC was the primary metric since it measures how well a model ranks customers by churn risk.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	CV-AUC
Logistic Regression ★	80.7%	65.0%	55.7%	60.0%	0.848	0.842
Random Forest	80.6%	64.4%	53.4%	58.3%	0.845	0.839
Gradient Boosting	80.0%	63.3%	53.4%	58.0%	0.844	0.837
Decision Tree	79.6%	60.2%	50.4%	54.7%	0.830	0.812

Selected model: Logistic Regression — highest ROC-AUC, confirmed by 5-fold cross-validation (0.842 ± 0.008). **Explainability:** permutation importance (model-agnostic, SHAP-equivalent — measures the AUC drop when a feature is shuffled) confirms **tenure**, **contract type**, and **monthly charges** as the strongest predictors, consistent with Random Forest / Gradient Boosting feature importances and the EDA findings above.

CUSTOMER SEGMENTATION — Every Customer Scored 0–100% and Grouped into 3 Segments

Segment	Threshold	Count	% of Base	Actual Churn	Monthly Revenue	Priority
At Risk	≥ 60%	976	13.9%	72.5%	$79,650	URGENT
Dormant	30–59%	1,609	22.8%	44.7%	$120,900	HIGH
Loyal	< 30%	4,458	63.3%	9.9%	$255,567	MAINTAIN

Chart 1 — Customer Segment Distribution

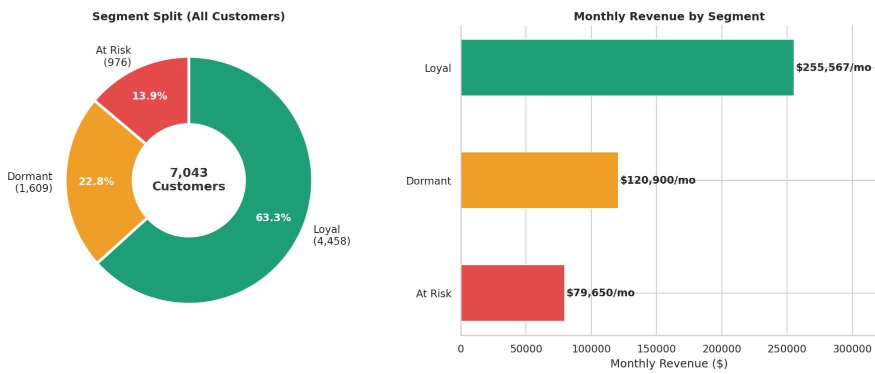

Chart — Customer Segment Distribution (count share & monthly revenue by segment)

BUSINESS RECOMMENDATIONS

AT RISK

976 cust. · $79,650/mo · URGENT

■ Personal outreach within 48 hrs to every customer with churn probability > 60%.

■ 20% discount, first 3 months, on a 1-year contract upgrade — converting 30% saves ~$23,900/mo.

■ Free 3-month OnlineSecurity + TechSupport bundle; these cut churn by ~27 pts for non-subscribers.

DORMANT

1,609 cust. · $120,900/mo · HIGH

■ 3-email re-engagement sequence (value reminder, usage insights, time-limited offer).

■ $10–15 loyalty credit for a 6-month commitment; price-lock incentive to upgrade contracts.

■ Quarterly satisfaction surveys to catch dissatisfaction signals before month-12 churn.

LOYAL

4,458 cust. · $255,567/mo · MAINTAIN

■ VIP milestone rewards at 12/24/36 months tenure (bill credits, upgrades, exclusive pricing).

■ Referral programme: $20–30 bill credit per successful referral — near-zero CAC growth.

■ Cross-sell premium bundles to grow ARPU without adding churn risk.

CONCLUSION

Contract type is the dominant churn driver — a 15× gap between month-to-month and two-year customers — with internet service type, payment method, and early tenure as secondary but significant drivers, confirmed by both EDA and model explainability. The validated Logistic Regression model (ROC-AUC 0.848, CV-AUC 0.842 ± 0.008) flags the 976 At-Risk customers responsible for $79,650 in monthly recurring revenue; retaining just 30% of them saves roughly **$23,900/month**, a return that comfortably justifies a targeted campaign. **Next step:** operationalise the model into a monthly scoring pipeline integrated with the CRM for automated, threshold-triggered outreach — closing the loop between data science and retention execution.